

CUSTOMER TARGET

Tunajenga Construction Projects

DEPLOYMENT SCOPE

DigitalOcean Cloud + Docker

PRIMARY USERS

Field Engineers, PMs, CAD Teams

USE CASE

Engineering Copilot & Compliance

1. PROBLEM STATEMENT

Tunajenga Company Limited executes multi-disciplinary civil engineering, structural construction, and infrastructure projects across diverse job sites. Field engineers, project managers, and quality control personnel rely on complex structural blueprints, building codes, safety SOPs, and site execution runbooks.

When unexpected field anomalies occur—such as foundation soil variations or concrete honeycomb defects—engineers spend critical time searching through heavy documentation. Standard keyword searches fail to verify whether retrieved guidance applies to specific site constraints, while generic AI tools risk hallucinatory recommendations that lead to catastrophic structural rework and safety non-compliance.

Core Business Challenge:

Field personnel need instantaneous, verifiable engineering guidance. Tunajenga requires all site recommendations to remain strictly grounded (\$IsSUP\$) and useful (\$IsUSE\$), relying on approved company blueprints and civil codes, with controlled external standards fallback when local SOPs are incomplete.

2. WHY TRADITIONAL RETRIEVAL IS INSUFFICIENT

Traditional Gap	Impact on Tunajenga Construction Sites
Static Blueprint Retrieval	Retrieves unverified drawing chunks without evaluating if context fits specific structural loads.
No Grounding Support Check	Generates unverified troubleshooting advice that risks building code violations and structural failure.
Isolated Project Knowledge	Site engineers lose conversation thread context when switching between complex inspection steps.
Lack of Standards Fallback	Fails when internal runbooks lack coverage for novel civil engineering techniques or revised local codes.

3. PROPOSED SOLUTION: SELF-RAG ENGINEERING COPILOT

Deploy an Enterprise Engineering Copilot powered by LangGraph, Pinecone, and OpenAI. The copilot restores conversation memory from SQLite, contextualizes site queries, retrieves grounded blueprints, grades evidence sufficiency (\$IsSUP\$), rewrites queries, and falls back to Tavily civil standards search only when internal sources are insufficient.

SITE QUERY DELAY

-65%

Reduction in engineering answer latency on job sites.

STRUCTURAL REWORK

-28%

Decrease in quality rework and non-compliance errors.

TRACEABILITY

100%

Full audit logging of source blueprints and building codes.

4. KEY ACCEPTANCE CRITERIA

Acceptance Test	Expected Result
Data Ingestion	data_ingestion.py processes site blueprints, CAD files, and SOPs into Pinecone tunajenga-kb.
Private KB Retrieval	Rebar spacing queries retrieve company SOPs and bypass external web search when evidence is complete.
Persistent Memory	Multi-step site inspection questions reuse past thread context restored from SQLite storage.
Standards Fallback	Novel civil code questions missing from private SOPs automatically trigger Tavily web search fallback.
Dynamic Ingestion	Uploading new site drawings via the web UI immediately indexes them for active field queries.